

Experiment-7

I-V Characteristics of pMOSFET

Name: S.Vathsalya

Roll.No:se25mvls020

Course :M.Tech VLSI &ES

Aim: Design p-channel Metal Oxide Semiconductor Field Effect Transistor (pMOSFET) and simulate the transfer (I_s-V_{sg}) and output (I_s-V_{sd}) characteristics of a using Synopsys TCAD Sentaurus.

Objectives:

- 1 Study the I_s-V_{sg} characteristics of p-MOSFET for different values of V_{sd} .
- 2 Study the I_s-V_{sd} characteristics of p-MOSFET for different values of V_{sg} .
- 3 Study the effect of device dimensions on the I_s-V_{sg} characteristics.
- 4 Study the effect of different materials on the I_s-V_{sg} characteristics.
- 5 Optimized the device structure for higher I_{on}/I_{off} ratio.

Software Tool:

Synopsys TCAD Sentaurus.

Theory: The metal–oxide–semiconductor field-effect transistor (MOSFET) is a transistor used for amplifying or switching electronic signals. In MOSFETs, a voltage on the oxide-insulated gate electrode can induce a conducting channel between the two other contacts called source and drain. The channel can be of n-type or p-type, and is accordingly called an pMOSFET or a nMOSFET.

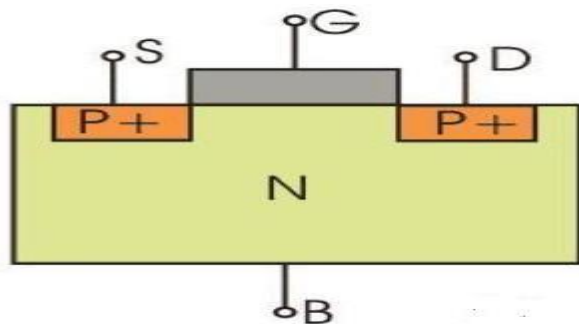


Fig :PMOSFET Structure

Procedure:

- 1.Create a new folder. Open Terminal in the new folder then enter “sde &” command in the terminal.
- 2.Go to View a GUI Mode a 2D Mode.
- 3.Go to Draw then deselect the “Auto Region Naming “and select the “Exact Coordinates”.
- 4.Now go to materials and select “silicon” (silicon is the default material) then click on “Create rectangular Sheet Region” on the toolbar.
- 5.Place the rectangle on Canvas with (0,0.045) and (0,0.180) coordinates and press ok.
- 6.Now enter Region Name as “substrate”.
- 7.Dope the substrate with Phosphorous Active concentration of $1e18$. For doping the silicon region, Go to Device a Constant Profile Placement.
- 8.Deposit oxide layer source on the substrate. Enter the region name as source.with

coordinates(0.00,0.025) and (0.0,0.06).

9.Dope the source with Boron Active concentration of $1e20$. For doping the silicon region, Go to Device a Constant Profile Placement.

10.Deposit oxide layer drain on the substrate. Enter the region name as drain.with coordinates(0.00,0.025) and (0.180,0.120).

11.Dope the Drain with Boron Active concentration of $1e19$. For doping the silicon region, Go to Device a Constant Profile Placement.

12..Deposit oxide layer hfo2 on the substrate. Enter the region name as oxide.with coordinates(0.00,0.060) and (-0.045,0.120).

13.Deposit metal layer on oxide layer. Choose GatePolySi as metal. Enter the region name as metal with coordinates (-0.0045,-0.009) and (0.060,0.120).

14.Give the contacts to gate source drain body. For creating Contact, Go to Contacts a Contact Sets.Enter the contact name as gatec ,bodyc,drainc,sourcec with different edge colour.

15.To Set Contact first change the Select Body to Select Edge on the Tool Bar.

16.Now click on Arrow () placed at the left panel and go to Contacts a Set Contact then place the electrical contact on the left and right edges of silicon.

17.Go to Mesh a Define Rel/Eval Window and select rectangle à draw rectangle on canvas covering the entire design.

18.Got Mesh à Refinement Placement enter the minimum & maximum element size. Click on create Refinement.

19.Go to Mesh à click on Build Mesh. Then the device structure will be open in the svisual.

20.Got Mesh à Refinement Placement enter the minimum & maximum element size.minimum size is $1e-06$ and 0.005 is the maximum element size and Click on create Refinement.

21.Go to Mesh click on Build Mesh. Then the device structure will be open in svisual.

22.To study the device characteristics, edit the pmos “cmd file” (.cmd file). msh.tdr file name should be included in cmd file Grid section.

23. Edit .ac cmd file for different frequencies and sweep voltages from 0V to -2V.

24.Enter the “sdevice cmdfilename” (Ex: sdevice TEST_ac.cmd) in working directory terminal.

25.To visualize the device characteristics, enter svisual command in terminal.

26.Go to file in svisual and open plot file.

27.For plotting I_s-V_{sg} characteristics select gatec outer voltage to x-axis and sourcec total current to left y-axis.

RESULT:

Fig 1a: I_s – V_{SG} characteristics of p-MOSFET for different values of V_{SD} Ion

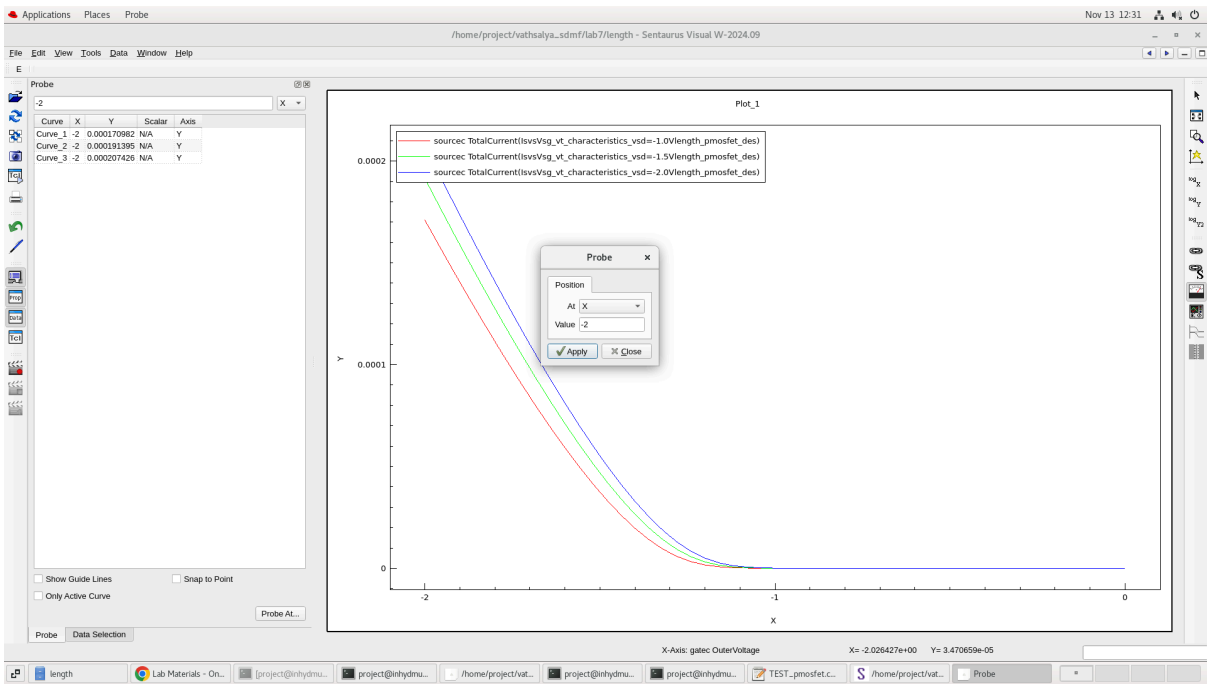


Fig 1b: I_s – V_{SG} characteristics of p-MOSFET for different values of V_{SD} Ioff

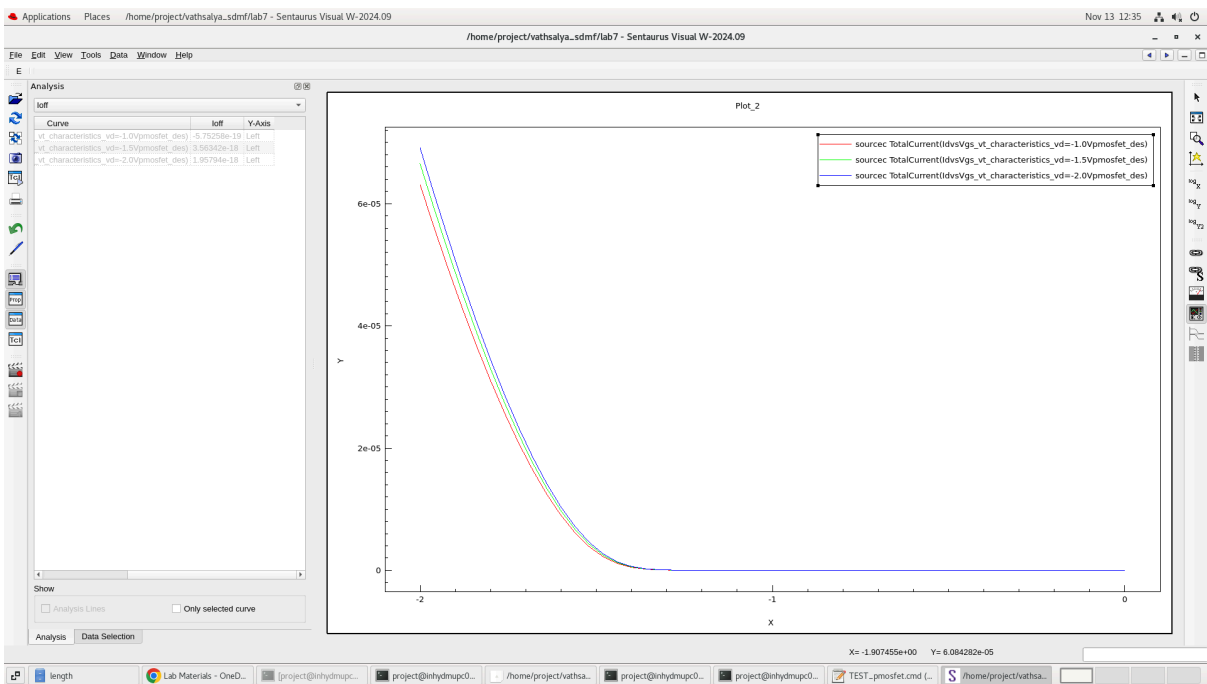


Fig 2: I_S - V_{SD} characteristics of p-MOSFET for different values of V_{SG} .

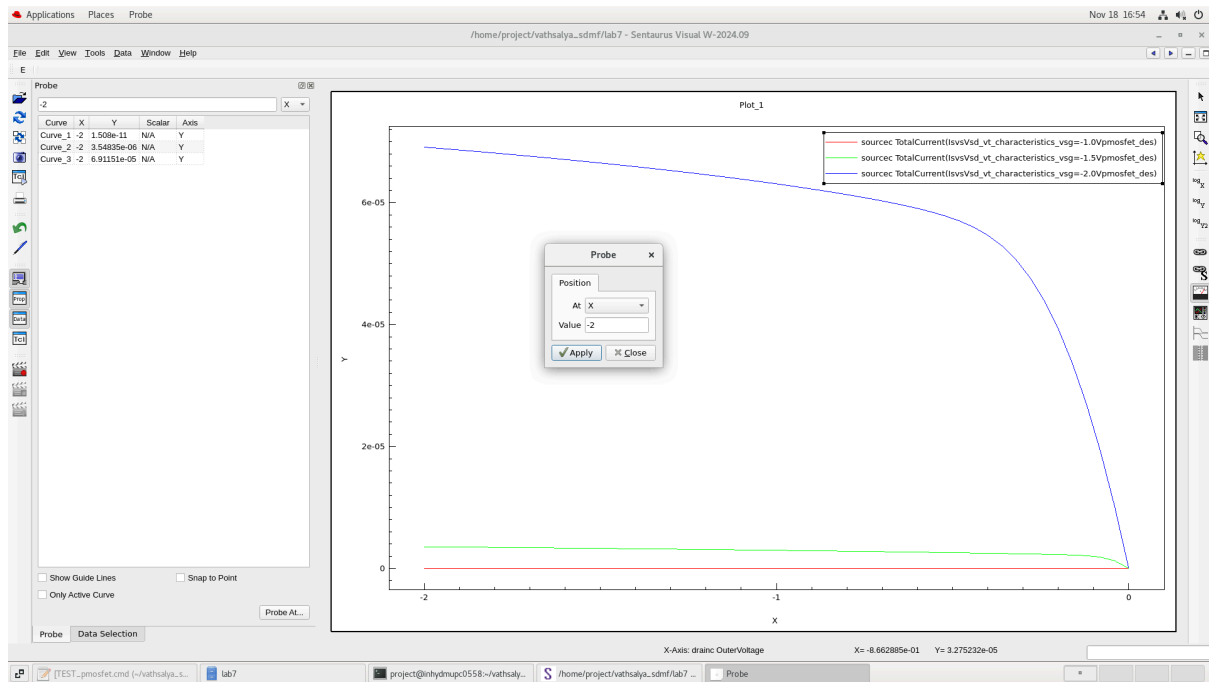


Fig 3a: device dimensions on the I_S - V_{SG} characteristics I_{on}
THE DIMENSIONS ARE REDUCED FROM $0.120\mu m$ to $0.06\mu m$

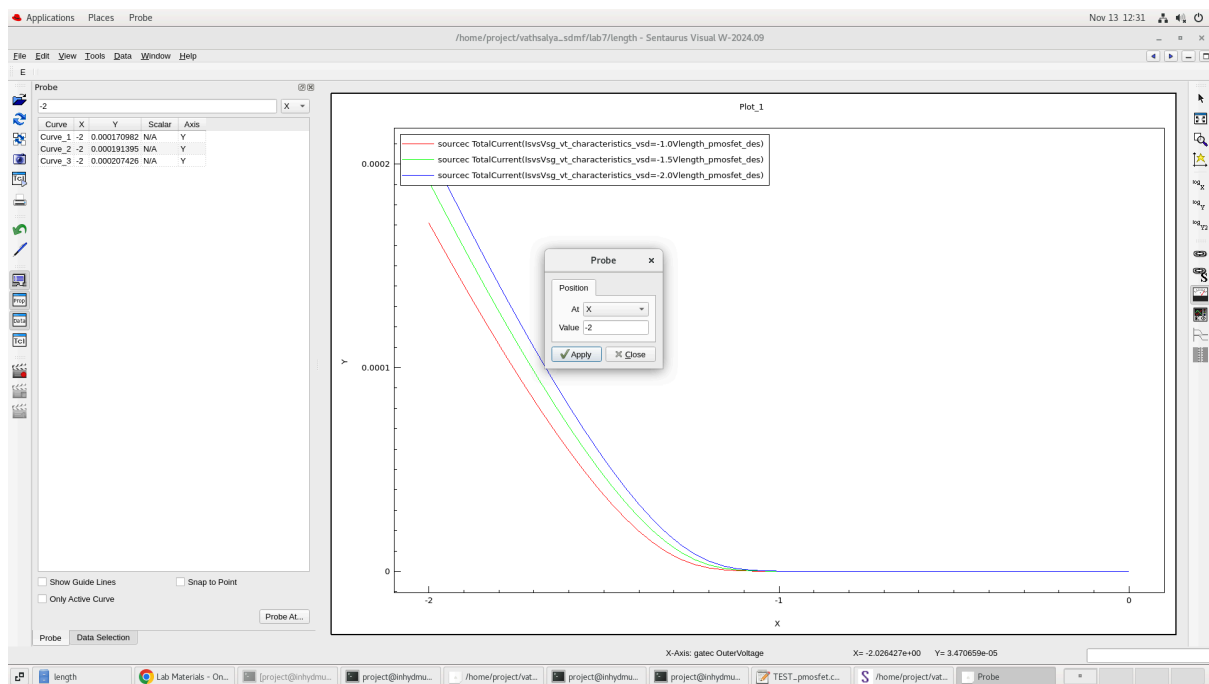


Fig 3b: Ioff

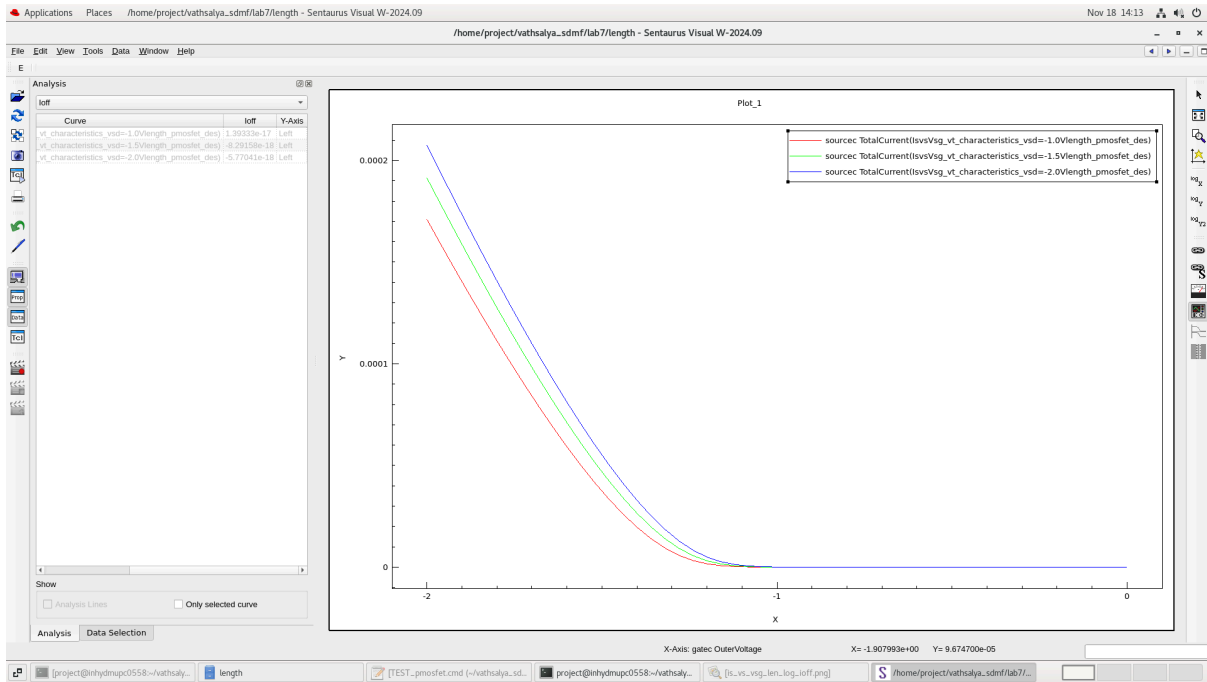


Fig 3c: Mesh of pmos for reduced length

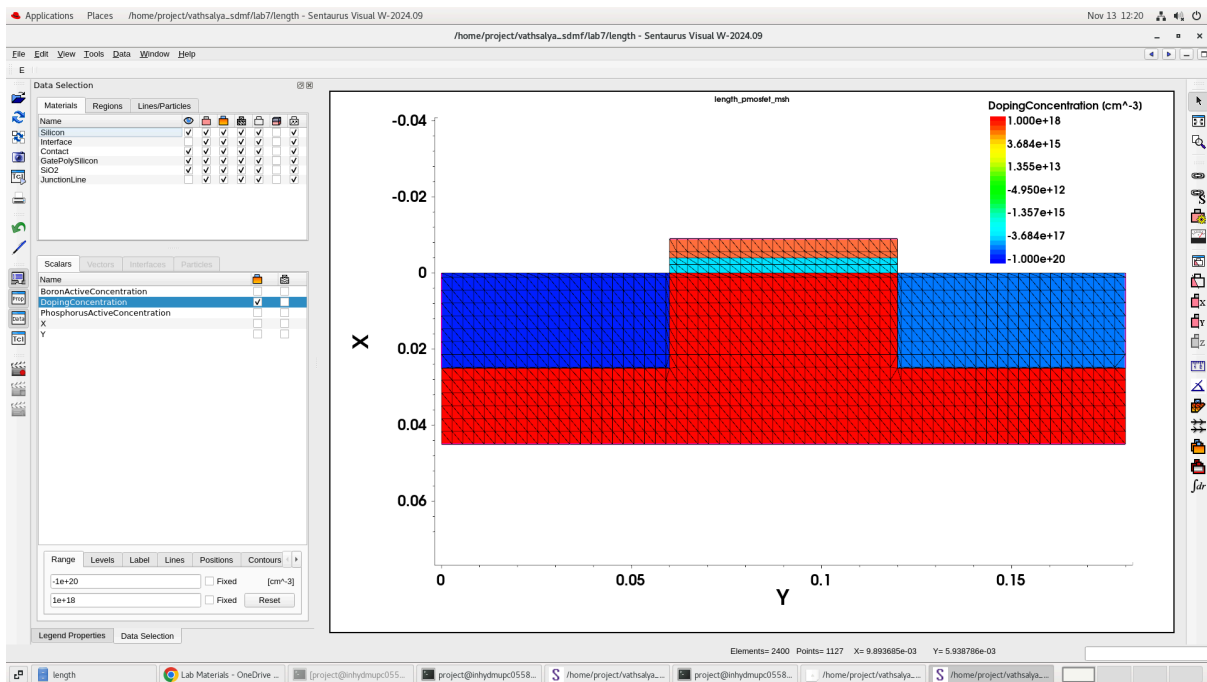


Fig 4a: Effect of GatePolySi on the I_s-V_{SG} characteristics Ion

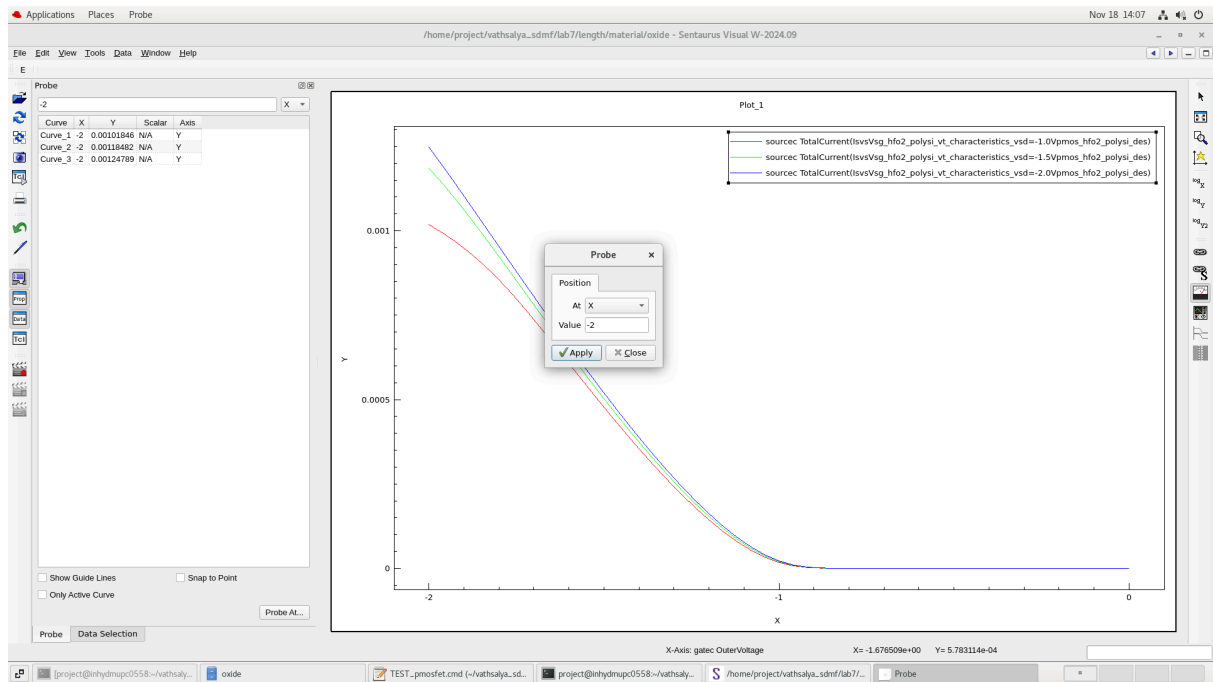


Fig 4b: Effect of GatePolySi on the I_s-V_{SG} characteristics Ioff

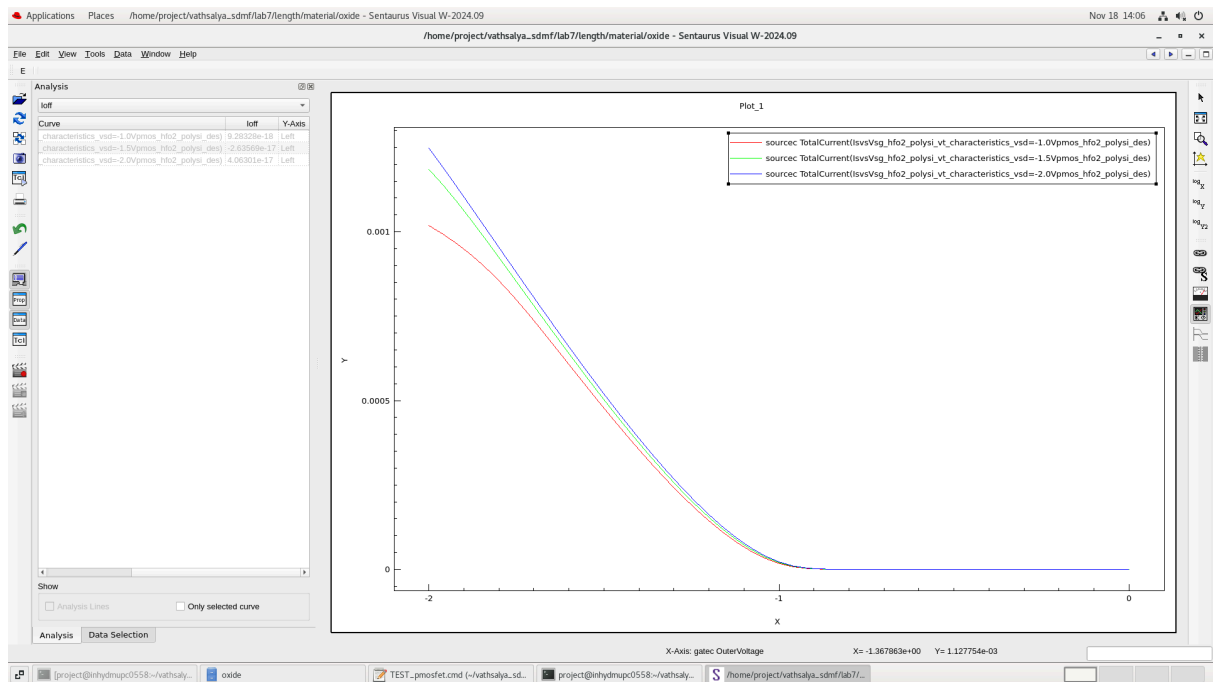


Fig 4c: Mesh of pmos for gate material as Gate PolySi

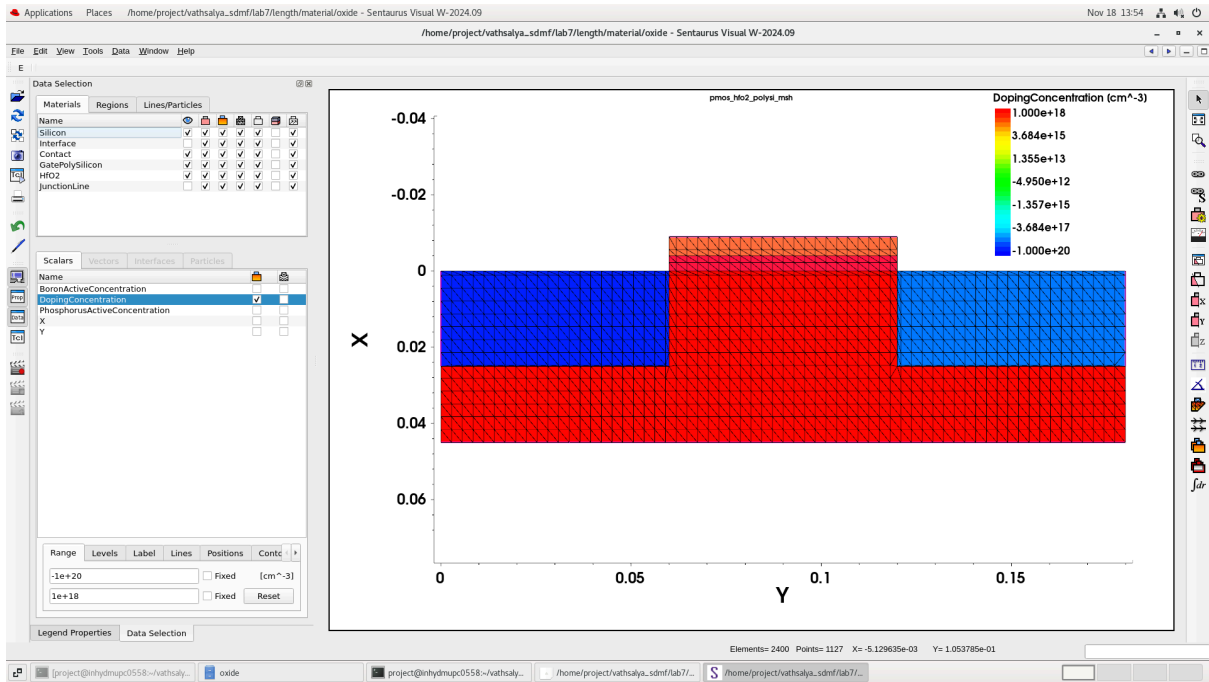


Fig 5a: Effect of Molybdenum on the I_S – V_{SG} characteristics Ion

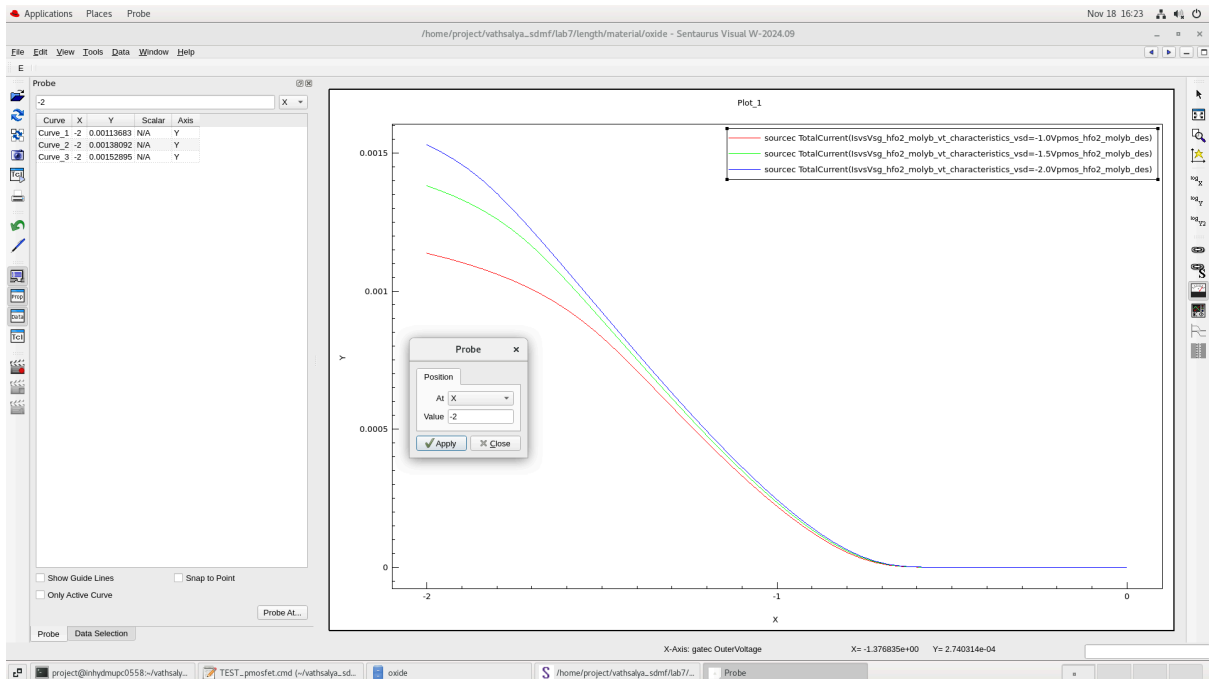


Fig 5b: Effect of Molybdenum on the I_S-V_{SG} characteristics Ioff

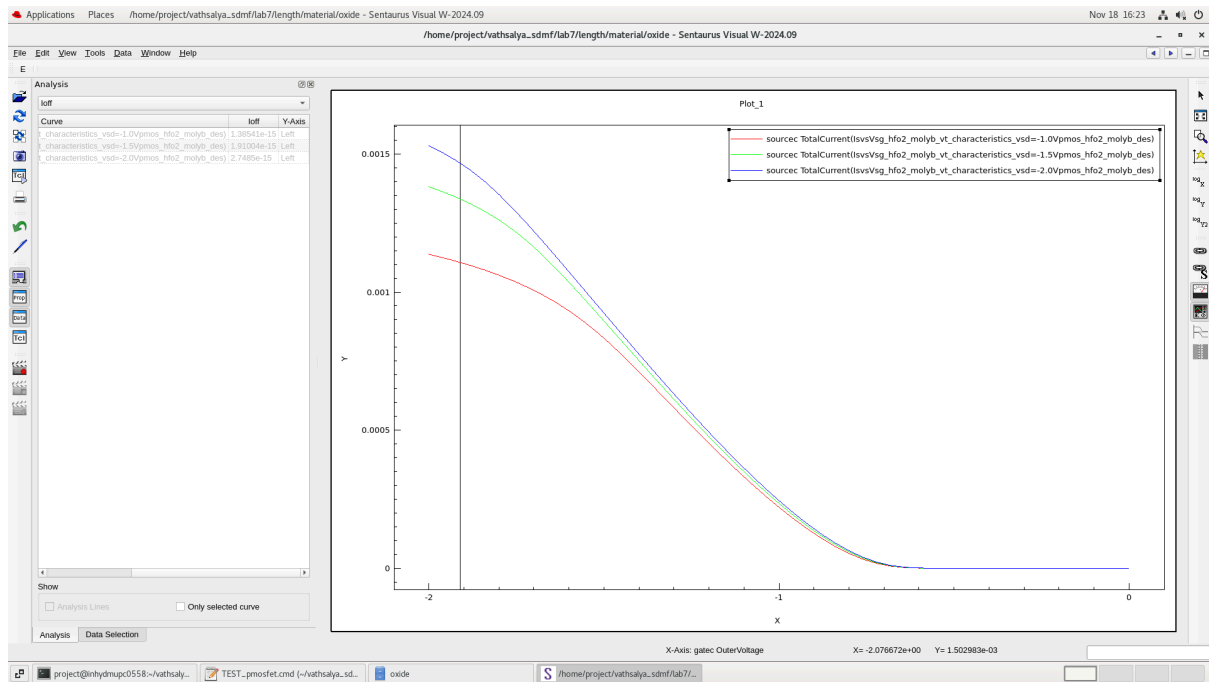


Fig 5c: Mesh of pmos for gate material as Molybdenum

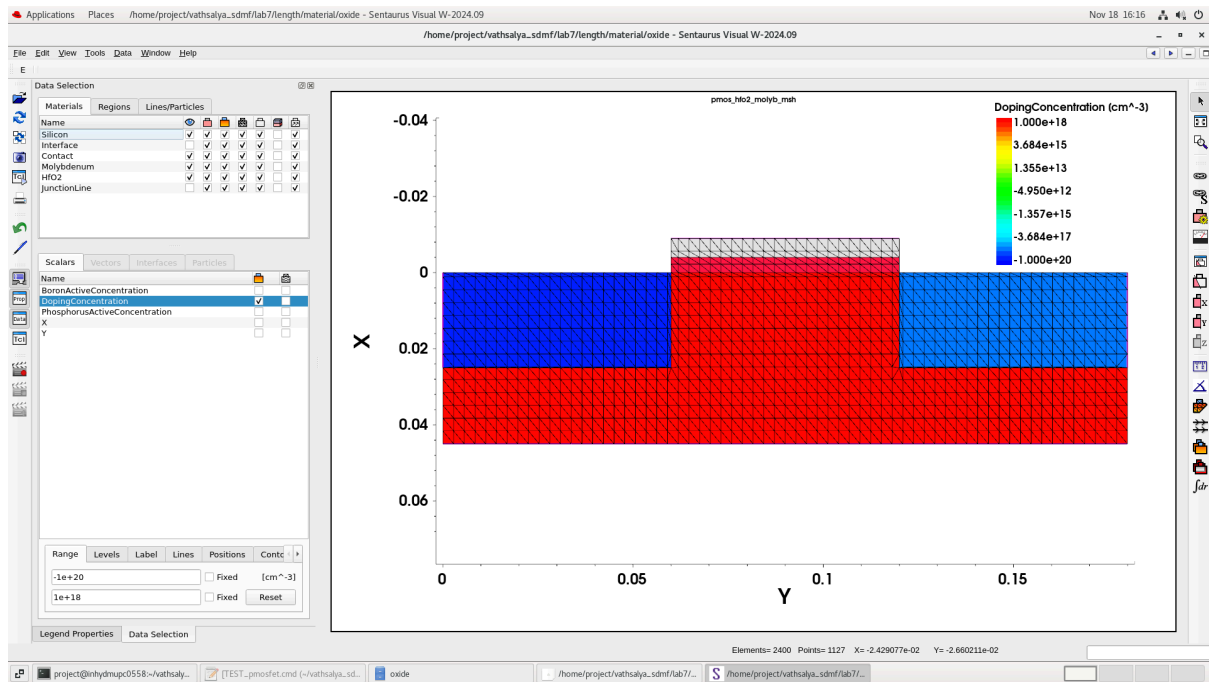


Fig 6a: Effect of Tin on the I_s – V_{SG} characteristics Ion

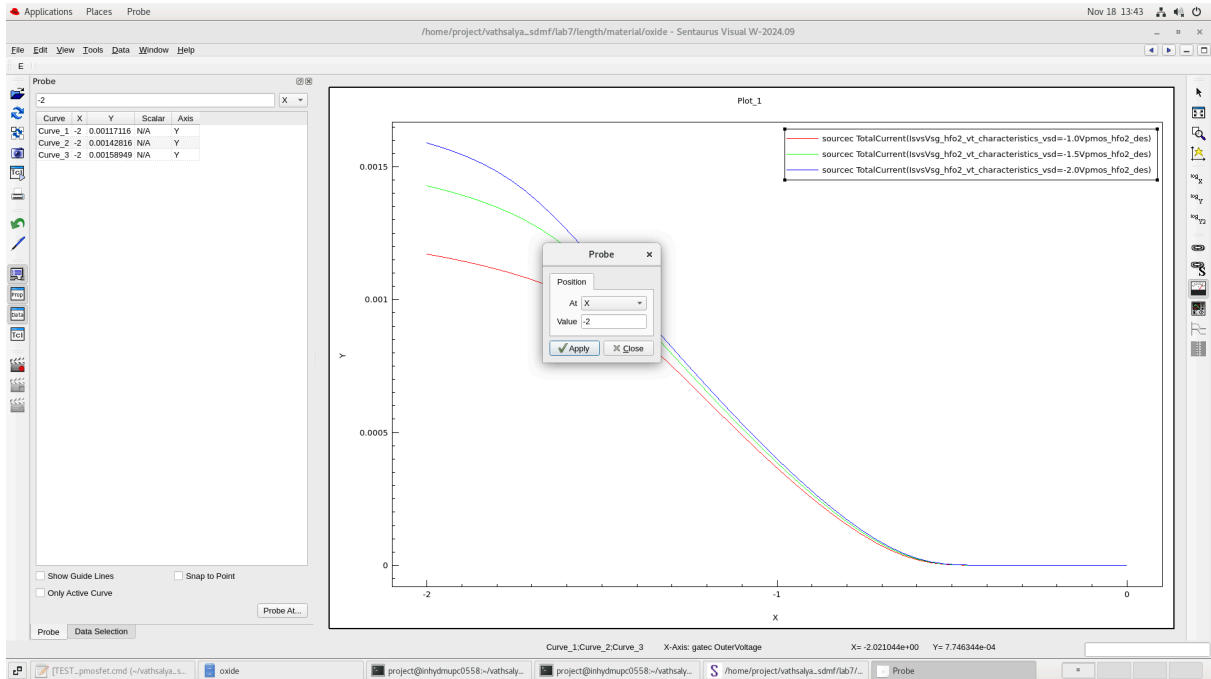


Fig 6b: Effect of Tin on the I_s – V_{SG} characteristics Ioff

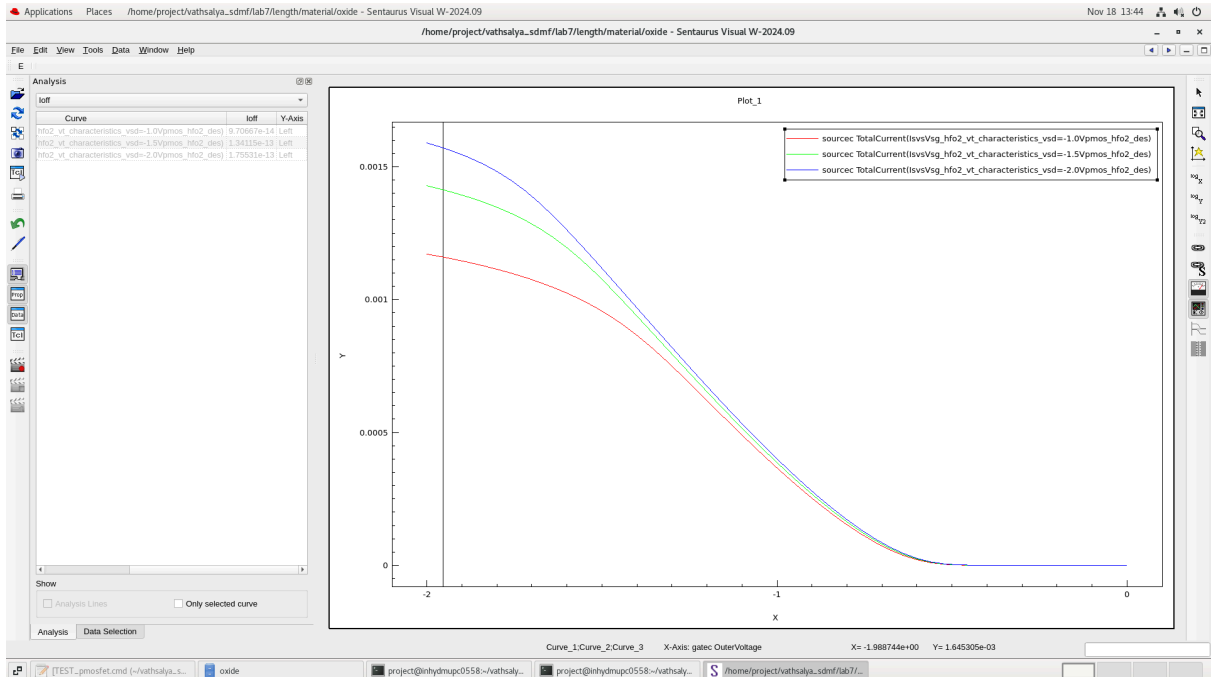
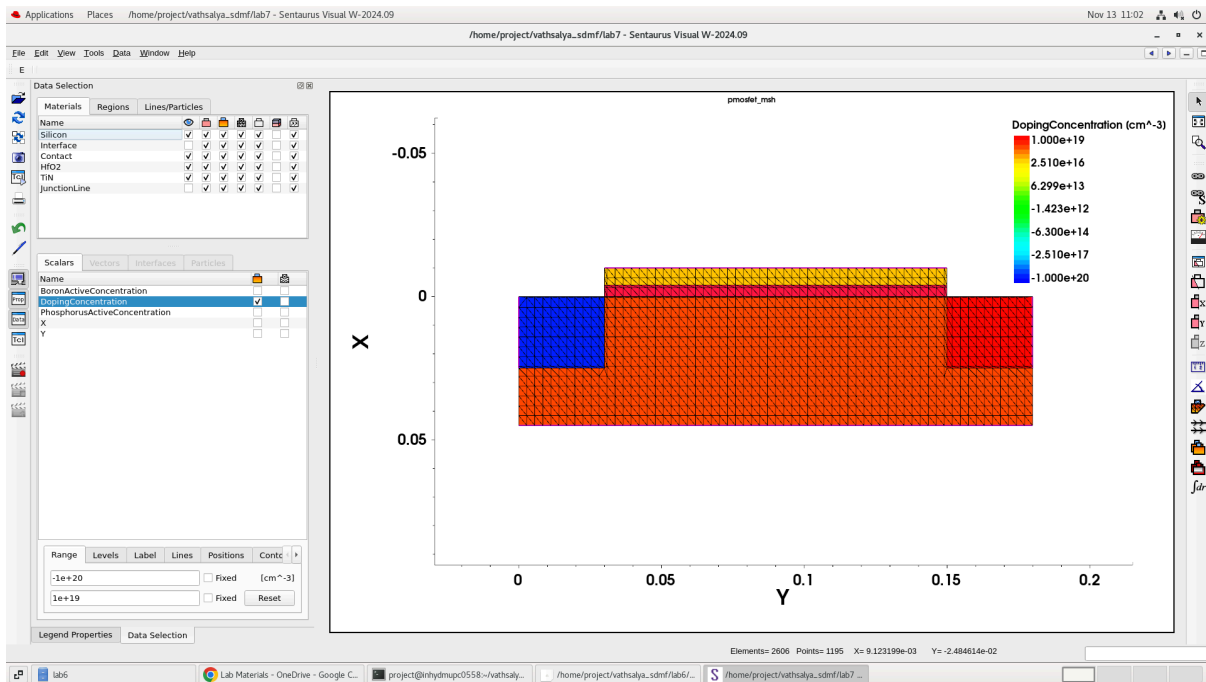


Fig 6c: Mesh of pmos for gate material as Tin



OBSERVATIONS:

1. As VSG increases, the source current I_S increases for all values of VSD because the p-channel becomes more strongly inverted.

When VSG is below the threshold, the current is almost zero (device in cut-off).

2. For small VSD, **I_S increases almost linearly** with VSD $\rightarrow$ ohmic/linear region of the p-MOSFET.

When **$VSD \geq VSG - |V_{TH}|$** , current enters **saturation**, and I_S becomes nearly constant.

3. When the **channel length (L) is reduced**, resistance decreases, so **I_S increases** for the same VSG.

Since resistance $\propto L$, a smaller L gives higher current and a steeper I_S -VSG curve.

4.1. Tin (Sn) – Highest Work Function (~ 4.7 – 4.8 eV)

Highest $|V_{TH}|$ (magnitude of threshold voltage)

Channel forms later (requires more negative VSG)

I_S starts increasing at higher VSG

Gives the lowest I_S (weakest conduction)

2. Molybdenum (Mo) – Medium Work Function (~ 4.3 – 4.5 eV)

Medium $|V_{TH}|$

Channel forms at moderate VSG

Moderate I_S

Curve lies between Sn and Polysilicon

3. Polysilicon Gate – Lowest Work Function (~ 4.05 eV)

Lowest $|V_{TH}|$

Channel forms early

IS rises quickly even at smaller VSG

Gives the highest IS

5.Optimised ion/ioff

For TiN

$V_{SD} = -1.0V$

$$\frac{I_{ON}}{I_{OFF}} = \frac{0.00117116}{9.70676E-14} = 1.287 \times 10^{10}$$

$V_{SD} = -1.5V$

$$\frac{I_{ON}}{I_{OFF}} = \frac{0.00148816}{1.34136E-13} = 1.084 \times 10^{10}$$

$V_{SD} = -2.0V$

$$\frac{I_{ON}}{I_{OFF}} = \frac{0.00158949}{1.75531E-13} = 9.06 \times 10^9$$

For PolySi Gate

$V_{SD} = -1.0V$

$$\frac{I_{ON}}{I_{OFF}} = \frac{0.00101846}{9.28328E-18} = 1.097 \times 10^{14}$$

$V_{SD} = -1.5V$

$$\frac{I_{ON}}{I_{OFF}} = \frac{0.00118482}{2.63560E-17} = 4.49 \times 10^{13}$$

$V_{SD} = -2.0V$

$$\frac{I_{ON}}{I_{OFF}} = \frac{0.00124789}{4.06301E-17} = 3.07 \times 10^{13}$$

For Molybdenum

$V_{SD} = -1.0V$

$$\frac{I_{ON}}{I_{OFF}} = \frac{0.00113683}{1.38514E-15} = 8.21 \times 10^{11}$$

$V_{SD} = -1.5V$

$$\frac{I_{ON}}{I_{OFF}} = \frac{0.00138092}{1.91004E-15} = 7.22 \times 10^{11}$$

$V_{SD} = -2.0V$

$$\frac{I_{ON}}{I_{OFF}} = \frac{0.00152895}{2.74856E-15} = 5.56 \times 10^{11}$$

Higher work function $\rightarrow$ higher $|V_{TH}| \rightarrow$ lower $I_{ON} \rightarrow$ lower I_{ON}/I_{OFF} in PMOS.

Conclusion: The transfer (I_S-V_{SG}) and output (I_S-V_{SD}) characteristics of an n-channel Metal Semiconductor Field Effect Transistor (pMOSFET) are studied using Synopsys TCAD Sentaurus software.

